# Radar range and velocity

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Separate a radar's range measurement from its line-of-sight velocity measurement.

**SUCCESS CRITERIA - I can...**

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Two EM waves in vacuum have frequencies f and 3 f. Compare speeds and wavelengths.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** range = $c\Delta t/2$; $c = 3.0\times10^{8}$ m/s.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

**Worked reasoning:** A radar echo returns after 2 microseconds. With $c=3\times10^{8}$ m/s, range$=c\Delta t/2=300$ m. The measured time includes the outward and return paths.

**Your turn:** Draw the round trip and label the conversion 2 microseconds$=2\times10^{-6}$ s.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

Draw outgoing and returning radar paths. Which measurement helps range: echo delay or frequency shift? Which helps radial velocity?




**Marzano check - circle:** ① ② ③

## 04 Build the solution

In a toy model, signal speed 20 m/s and echo delay 3 s. Find total path and range.

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Use the idea

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary? Sketch another possible motion.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A wave has frequency 5 Hz and wavelength 7 m. Find its speed.

**Marzano check - circle:** 1 2 3

### 07 Exit check

An echo’s out-and-back distance represents 1000 m. How far is the target?

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

Echo times change from 4 to 3 microseconds. Calculate both ranges and state whether the target is approaching or receding.

**Marzano check - circle:** 1 2 3